

Администрирование локальных сетей

Лабораторная работа № 9. Использование протокола STP. Агрегирование каналов

Абдуллахи Шугофа

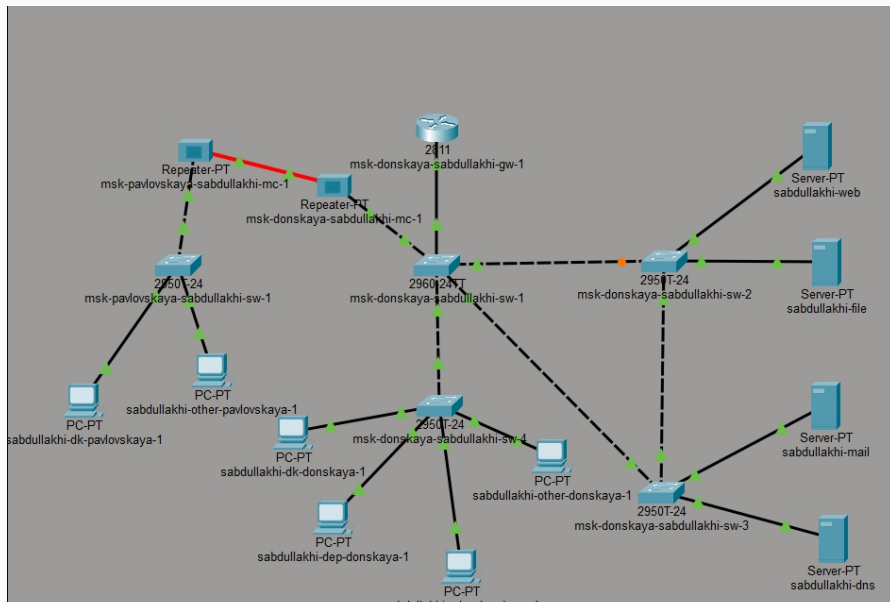
10 Апреля 2026

Российский университет дружбы народов, Москва, Россия

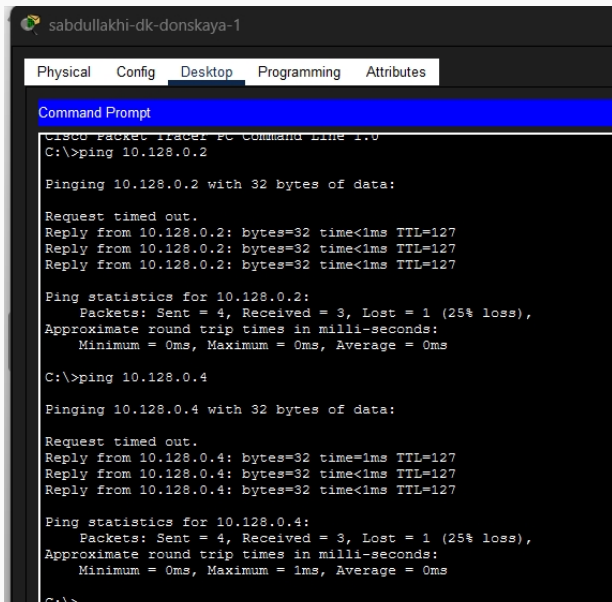
1.Цель работы

Изучение возможностей протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.

Изменение топологии и создание резервного соединения



Проверка прохождения ICMP-пакетов



The screenshot shows a Cisco Packet Tracer interface with a Command Prompt window open. The window title is 'Command Prompt' and the user is 'sabdullakhi-dk-donskaya-1'. The tabs at the top are 'Physical', 'Config', 'Desktop' (selected), 'Programming', and 'Attributes'. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

движение ICMP-пакетов

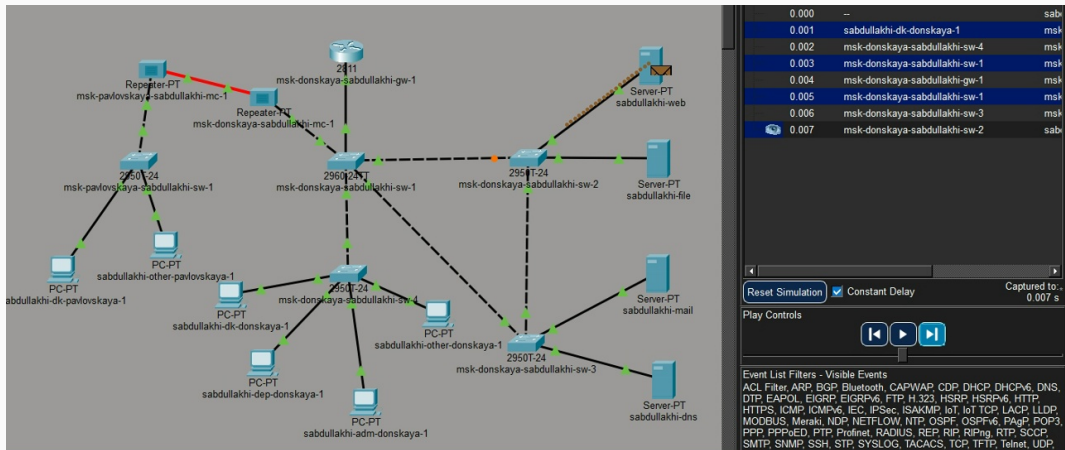


Рис. 3: Движение ICMP-пакетов через msk-donskaya-sabdullakhi-sw-2

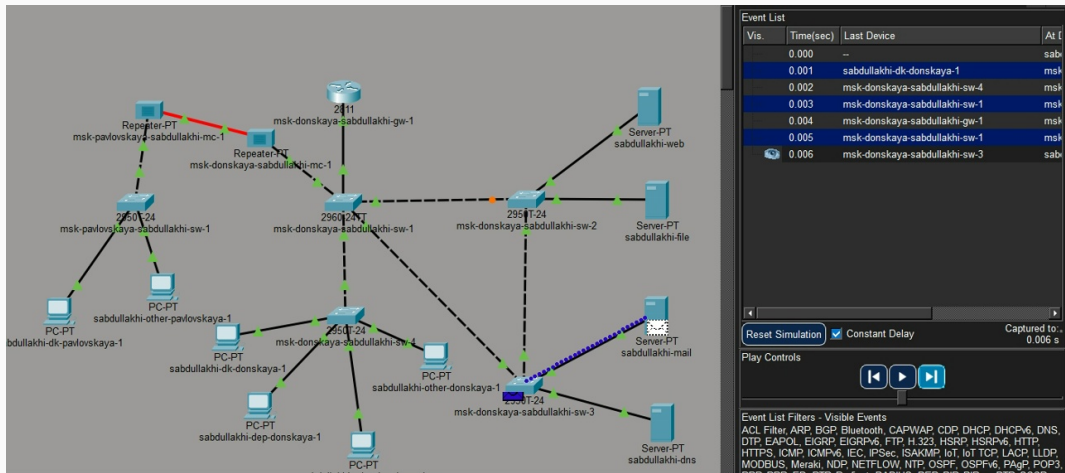


Рис. 4: Движение ICMP-пакетов через msk-donskaya-sabdullakhi-sw-4

Просмотр состояния протокола STP для VLAN 3

```
msk-donskaya-sabdullakhi-sw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-sabdullakhi-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-sabdullakhi-sw-1(config)#exit
msk-donskaya-sabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
msk-donskaya-sabdullakhi-sw-1#show spanning-tree vlan 4
```

No spanning tree instance exists.

```
msk-donskaya-sabdullakhi-sw-1#show spanning-tree vlan 3
VLAN0003
```

Spanning tree enabled protocol ieee

Root ID Priority 24579
 Address 0001.C7B0.C132

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

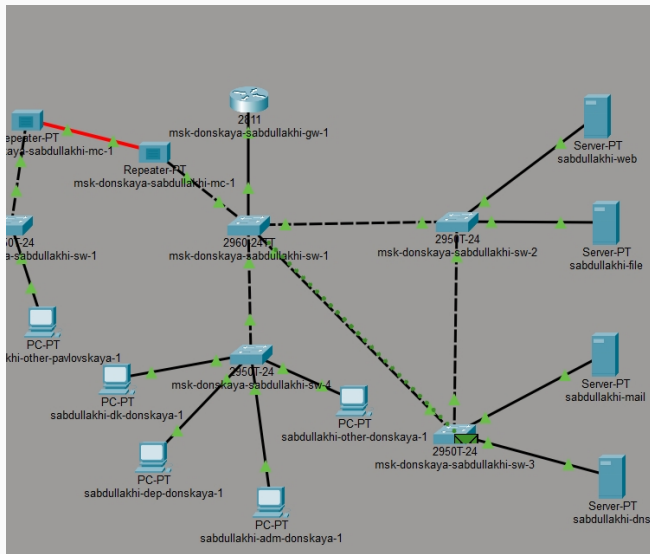
Bridge ID Priority 24579 (priority 24576 sys-id-ext 3)

Address 0001.C7B0.C132
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/24	Desg	FWD	19	128.24	P2p
Fa0/23	Desg	FWD	19	128.23	P2p
Gi0/2	Desg	FWD	4	128.26	P2p
Gi0/1	Desg	FWD	4	128.25	P2p

```
msk-donskaya-sabdullakhi-sw-1#
```

Изменение маршрута ICMP-пакетов после смены корневого моста



Event List

Vis.	Time(sec)	Last Device	At
	0.000	--	sabdullakhi-dk-donskaya-1
	0.001	sabdullakhi-dk-donskaya-1	msk-donskaya-sabdullakhi-sw-1
	0.002	msk-donskaya-sabdullakhi-sw-4	msk-donskaya-sabdullakhi-sw-1
	0.003	msk-donskaya-sabdullakhi-sw-1	msk-donskaya-sabdullakhi-gw-1
	0.004	msk-donskaya-sabdullakhi-gw-1	msk-donskaya-sabdullakhi-sw-1
	0.005	msk-donskaya-sabdullakhi-sw-1	msk-donskaya-sabdullakhi-sw-1

Reset Simulation ☒ Constant Delay Captured to: 0.005 s

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP,

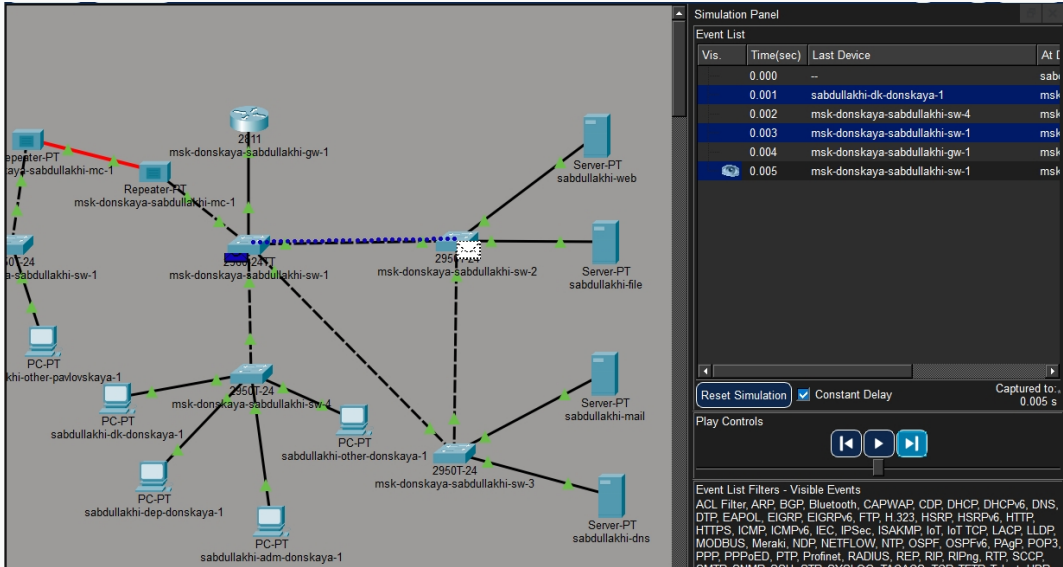


Рис. 7: Топология sabdullakhi-web

Настройка режима PortFast на серверных портах

```
msk-donskaya-sabdullakhi-sw-2#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-sabdullakhi-sw-2(config)#int f0/1
msk-donskaya-sabdullakhi-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-sabdullakhi-sw-2(config-if)#int f0/2
msk-donskaya-sabdullakhi-sw-2(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-sabdullakhi-sw-2(config-if)#
```

Настройка PortFast на msk-donskaya-sabdullakhi-sw-3

```
msk-donskaya-sabdullakhi-sw-3>ena
Password:
msk-donskaya-sabdullakhi-sw-3#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-sabdullakhi-sw-3(config)#int f0/1
msk-donskaya-sabdullakhi-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-sabdullakhi-sw-3(config-if)#int f0/2
msk-donskaya-sabdullakhi-sw-3(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
msk-donskaya-sabdullakhi-sw-3(config-if)#
```

Проверка отказоустойчивости классического STP

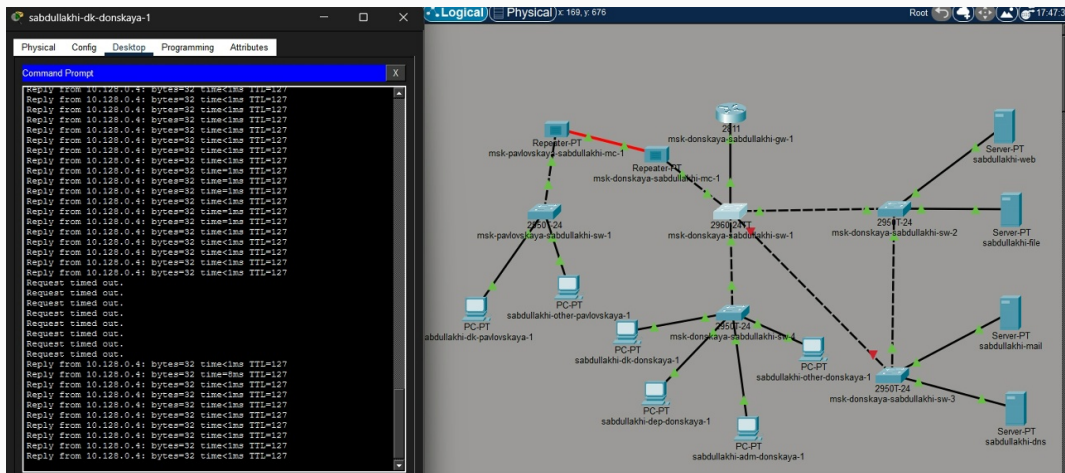


Рис. 10: Проверка отказоустойчивости STP

Перевод коммутаторов в режим Rapid PVST+

```
msk-donskaya-sabdullakhi-sw-1(config)#spanning-tree mode rapid-pvst
msk-donskaya-sabdullakhi-sw-1(config)#exit
msk-donskaya-sabdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-sabdullakhi-sw-1#wr mem
Building configuration...
[OK]
msk-donskaya-sabdullakhi-sw-1#
```

Рис. 11: Перевод sw-1 в режим Rapid PVST+

```
msk-donskaya-sabdullakhi-sw-2(config)#spanning-tree mode rapid-pvst
msk-donskaya-sabdullakhi-sw-2(config)#exit
msk-donskaya-sabdullakhi-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-sabdullakhi-sw-2#wr mem
Building configuration...
[OK]
msk-donskaya-sabdullakhi-sw-2#
```

Рис. 12: Перевод sw-2 в режим Rapid PVST+

Проверка отказоустойчивости Rapid PVST+

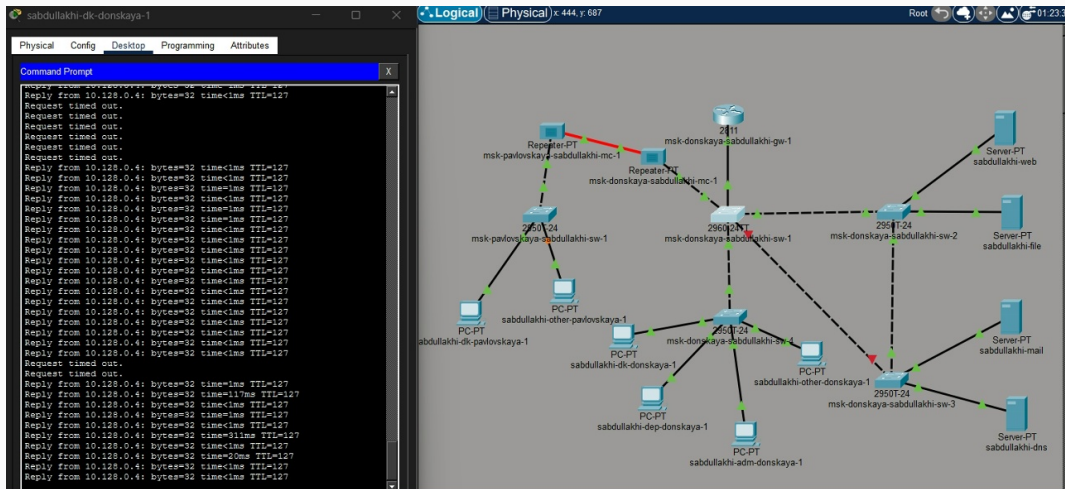


Рис. 13: Проверка работы сети при использовании Rapid PVST+

Настройка EtherChannel на коммутаторе msk-donskaya-sabdullakhi-sw-1

```
msk-donskaya-sabdullakhi-sw-1(config)#int range f0/20 - 23
msk-donskaya-sabdullakhi-sw-1(config-if-range)#channel-group 1 mode on
msk-donskaya-sabdullakhi-sw-1(config-if-range)#
%EC-5-CANNOT_BUNDLE2: Fa0/20 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/20 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/21 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/21 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/22 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/22 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of
Fa0/23 is full, Fa0/20 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of
Fa0/23 is full, Fa0/21 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of
Fa0/23 is full, Fa0/22 is half)

msk-donskaya-sabdullakhi-sw-1(config-if-range)#exit
msk-donskaya-sabdullakhi-sw-1(config)#int port-channel 1
msk-donskaya-sabdullakhi-sw-1(config-if)#sw mode trunk
msk-donskaya-sabdullakhi-sw-1(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/20 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/20 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/21 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/21 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/22 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/22 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of
Fa0/23 is full, Fa0/20 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of
Fa0/23 is full, Fa0/21 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of
Fa0/23 is full, Fa0/22 is half)

msk-donskaya-sabdullakhi-sw-1(config-if)#
```

Настройка EtherChannel на коммутаторе msk-donskaya-sabdullakhi-sw-4

```
msk-donskaya-sabdullakhi-sw-4(config-if-range)#no sw access vlan 104
msk-donskaya-sabdullakhi-sw-4(config-if-range)#exit
msk-donskaya-sabdullakhi-sw-4(config)#int range f0/20 - 23
msk-donskaya-sabdullakhi-sw-4(config-if-range)#channel-group 1 mode on
msk-donskaya-sabdullakhi-sw-4(config-if-range)#
Creating a port-channel interface Port-channel 1

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of
Fa0/23 is full, Fa0/20 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of
Fa0/23 is full, Fa0/21 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of
Fa0/23 is full, Fa0/22 is half)

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to down

msk-donskaya-sabdullakhi-sw-4(config-if-range)#exit
msk-donskaya-sabdullakhi-sw-4(config)#int port-channel 1
msk-donskaya-sabdullakhi-sw-4(config-if)#sw mode trunk
msk-donskaya-sabdullakhi-sw-4(config-if)#
%EC-5-CANNOT_BUNDLE2: Fa0/20 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/20 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/21 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/21 is half, Fa0/23 is full)

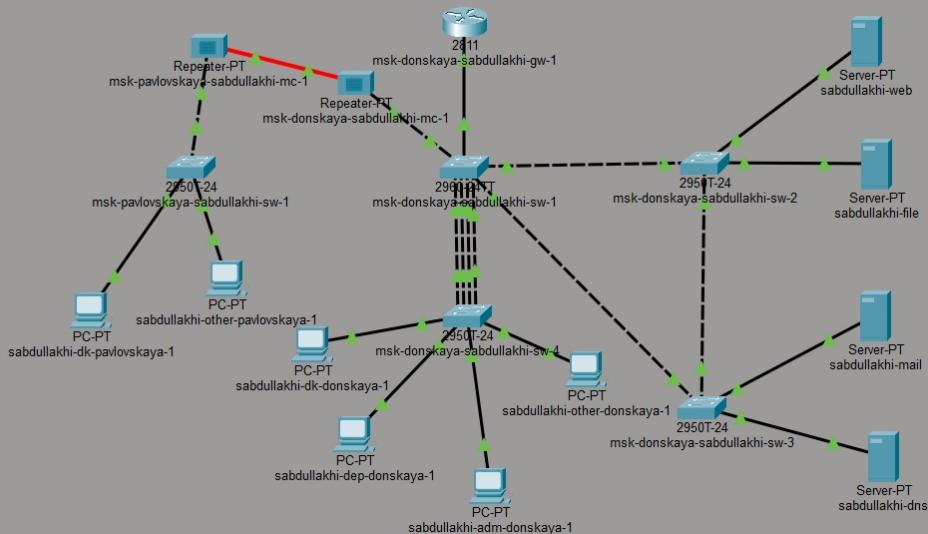
%EC-5-CANNOT_BUNDLE2: Fa0/22 is not compatible with Fa0/23 and will be suspended (duplex of
Fa0/22 is half, Fa0/23 is full)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (duplex of
Fa0/23 is full, Fa0/20 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (duplex of
Fa0/23 is full, Fa0/21 is half)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (duplex of
Fa0/23 is full, Fa0/22 is half)
```

Сформированное агрегированное соединение между коммутаторами



Успешная проверка соединения после настройки EtherChannel

```
C:\>ping -n 1000 mail.donskaya.rudn.ru
```

```
Pinging 10.128.0.4 with 32 bytes of data:
```

```
Reply from 10.128.0.4: bytes=32 time=10ms TTL=127
```

```
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
```

```
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
```

```
Reply from 10.128.0.4: bytes=32 time=7ms TTL=127
```

```
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 10.128.0.4:
```

```
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 10ms, Average = 3ms
```

```
Control-C
```

```
^C
```

```
C:\>
```

В ходе работы организовано резервное соединение между sw-1 и sw-3. Изучен протокол STP, проведена смена корневого коммутатора. На серверных портах настроен PortFast. Классический STP восстанавливается за 30-50 с, Rapid PVST+ — за 1-3 с. Настроен EtherChannel между sw-1 и sw-4. Построена отказоустойчивая сеть с быстрым восстановлением.

Спасибо за внимание
